

CIRCL
Computer Incident
Response Center
Luxembourg



AIL Project - Practical and Efficient Data-Mining of Chats, Suspicious Websites, Forums and Tor Hidden-Services

WARFARE

https://ail-project.org/ - HOPLITE

Aurelien Thirion - aurelien.thirion@circl.lu

April 22, 2026

CIRCL https://www.circl.lu

Background

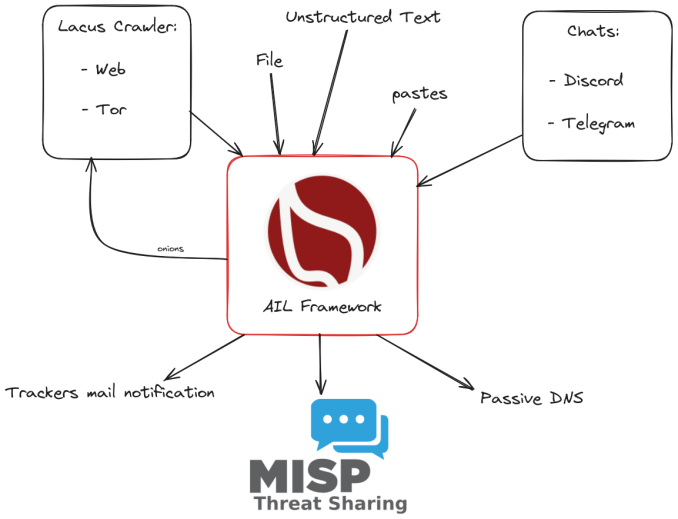
- Over the past years, CIRCL has developed the AIL project[1] to fulfill our needs at CIRCL in intelligence gathering and analysis.
- AIL features an extensible Python-based framework for the **analysis of unstructured information**, collected either through an advanced crawler manager or from various feeders, including social networks and custom feeders.
- The AIL Project is an **open-source** framework[2] comprising various modules designed for the **collection, crawling, digging, and analysis of unstructured data**.



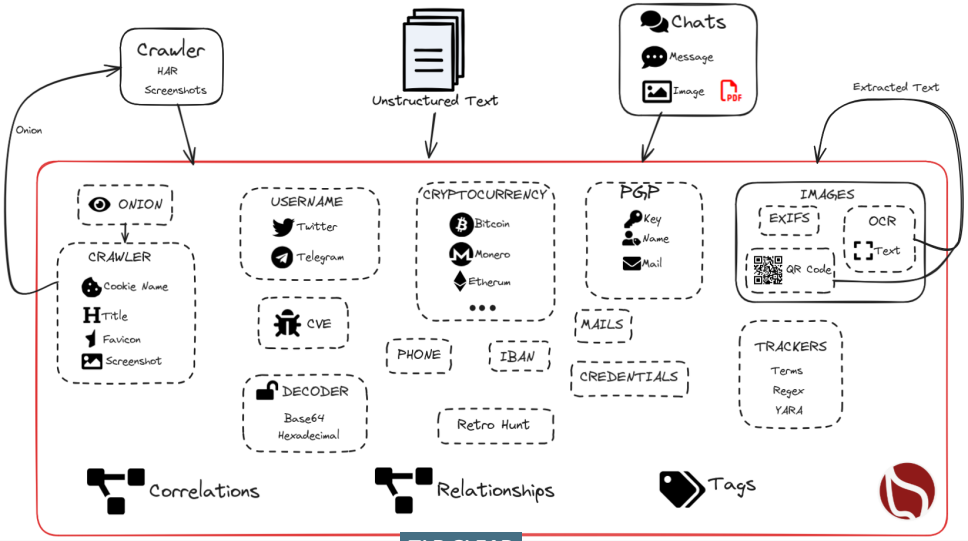
[1] https://www.ail-project.org/
[2] https://github.com/ail-project

Architecture

High Level Overview



Analysis of unstructured information



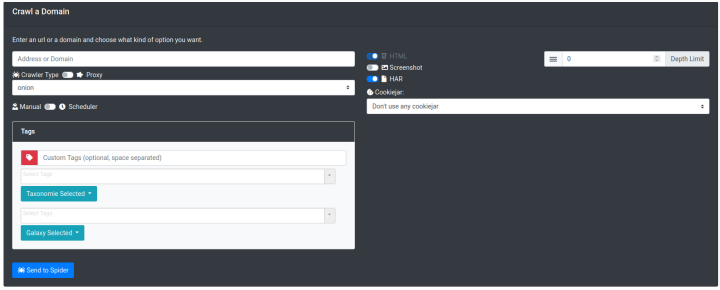
AIL Intelligence Lifecycle

- **Collection**
  - Continuous ingestion from chats, websites, files, ...
- **Processing**
  - Transform unstructured content into structured information
  - Extraction, decoding, and tagging
- **Detection**
  - Identify suspicious patterns using trackers and YARA rules
- **Analysis**
  - Correlate activity and investigate actors across platforms
- **Dissemination**
  - Share findings and export to MISP

Collection

Collection - Automate crawling

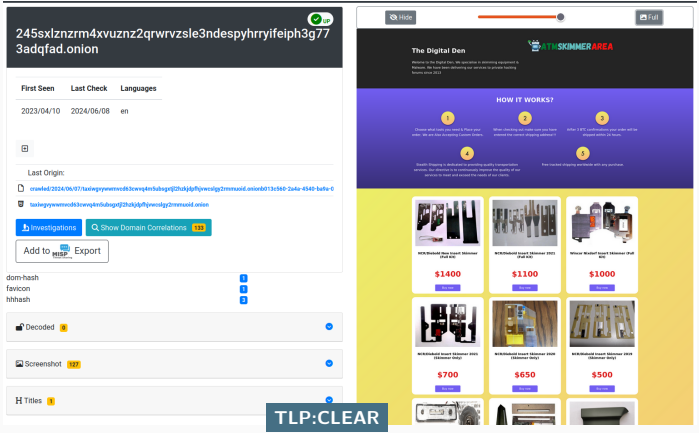
- Crawling can be a challenging task, for example, gathering all the blog posts from ransomware groups[3], which can be demanding for an analyst.
- AIL offers a crawling feature that can **initiate regular crawls using a standard spawned browser**.



[3] https://www.ransomlook.io/

Collection - Authenticated crawling

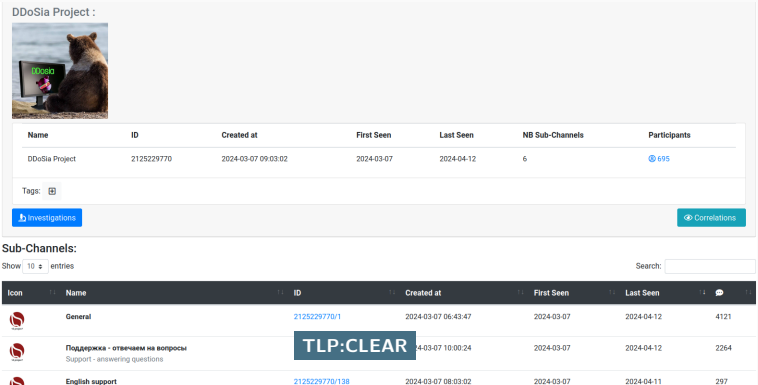
- Uses a real browser environment
- Supports authenticated sessions, handles cookies and local storage
- Continuously monitors sources over time

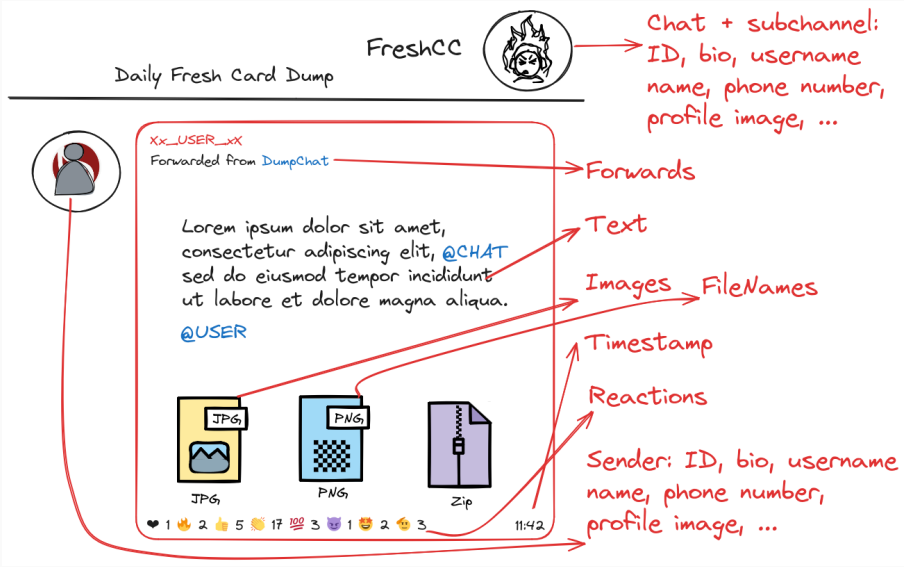


TLP:CLEAR

Collection - Automate Collection

- Collecting data from various chat sources can be a **tedious task for analysts**.
- AIL offers a set of feeders (e.g., Telegram, Discord) that can be used to subscribe to chat channels.
- All the **collected messages are then processed and analyzed** within AIL's *processing* and *analysis* stages.



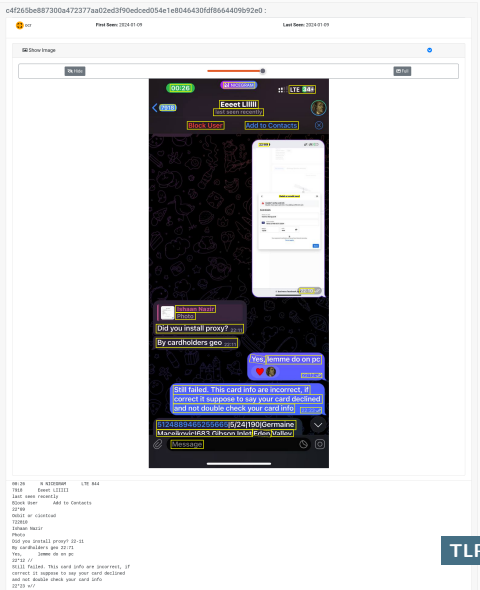

Daily Fresh Card Dump
FreshCC
Chat + subchannel:
ID, bio, username
name, phone number,
profile image, ...
Xx_USER_xX
Forwarded from DumpChat
Forwards
Lorem ipsum dolor sit amet,
consectetur adipiscing elit, @CHAT
sed do eiusmod tempor incididunt
ut labore et dolore magna aliqua.
@USER
Text
Images
FileNames
Timestamp
Reactions
JPG
PNG
JPG
PNG
Zip
❤ 1 🔥 2 👍 5 👏 17 💯 3 😈 1 🤩 2 🤔 3
11:42
Sender: ID, bio, username
name, phone number,
profile image, ...

Processing

Processing Unsctrutured content

- Collected data is often unstructured:
  - Free text, screenshots, documents, encoded data, ...
- Relevant information is often embedded within the content:
  - Identifiers inside images or files
  - Obfuscated or partially formatted data
- Direct search and correlation are unreliable without normalization

OCR: Optical Character Recognition

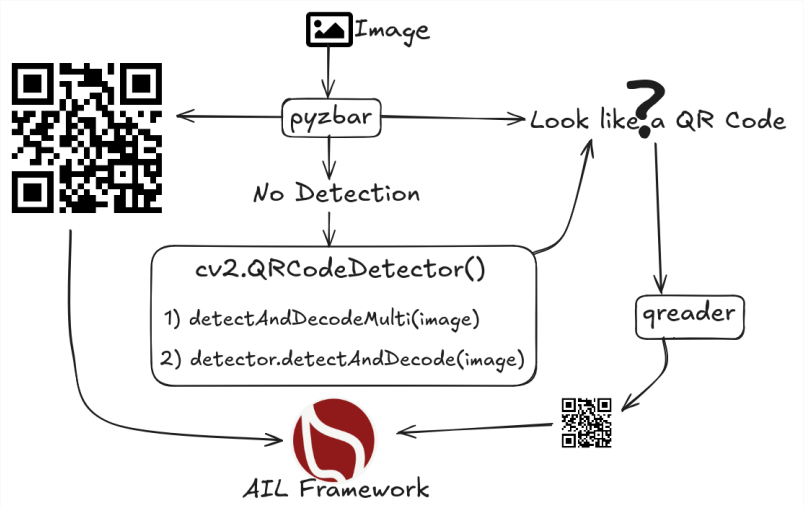


- Threat actors are often verbose and frequently **share extensive details in private channels**.
- Many messages contain screenshots and images.
- Text detection and extraction are performed across 80+ languages using a CRNN (Convolutional Recurrent Neural Network).
- **Enables keyword-based matching and detection**.

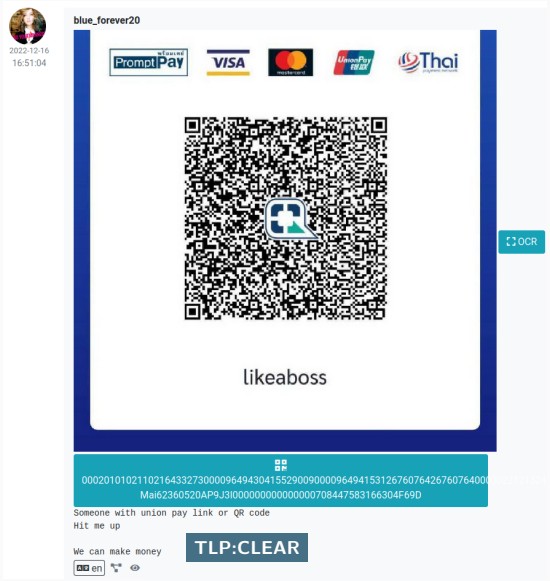
Reprocessing Extracted OCR Content

- Extracted text is processed like normal messages:
  - entity extraction (urls, usernames, cryptocurrencies, credentials, . . .)
  - tagging and classification
  - trackers and retro hunts on content that would otherwise be invisible

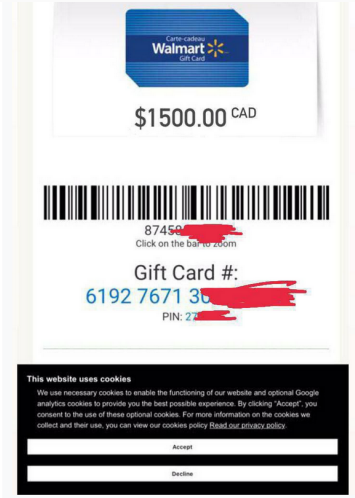
QR Code Extractor



QR Codes


blue_forever20
2022-12-16
16:51:04
PromptPay
VISA
UnionPay
Thai
OCR
likeaboss
Someone with union pay link or QR code
Hit me up
We can make money
en
TLP:CLEAR

Bar Codes



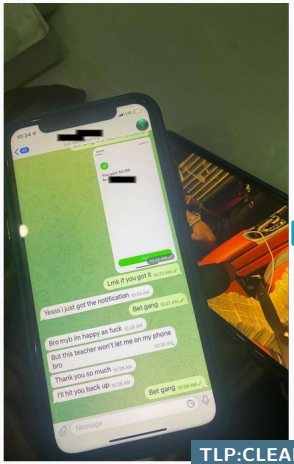
OCR 8745860537300061927671300649 13

Embedded Data Extraction

- QR codes and barcodes often embed hidden data:
  - urls, invite links, payment references
  - account identifiers, tracking IDs
  - encoded payloads shared in screenshots
- AIL extracts this content and reprocesses it
- Other modules decode and reprocess encoded data (Base64, . . .)

Images Descriptions

- Uses **Ollama** and **Qwen2.5-VL** to automatically generate descriptions for screenshots and other images.



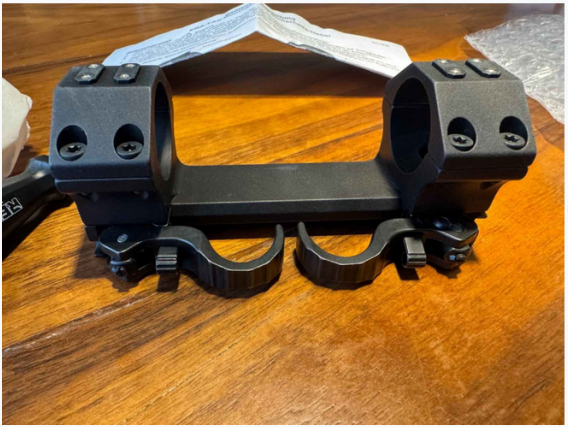
OCR

The picture shows a smartphone displaying a WhatsApp conversation. The conversation is between two people, with one person sending a message and the other responding. The message content appears to be about a financial transaction, with the sender mentioning sending $2.100 to someone named [redacted]. The conversation also includes some casual and informal language, with phrases like "Lmk if you got it," "Yesss i just got the notification," and "Bro myb im happy as fuck." The time stamps on the messages indicate that the conversation took place around 10:23 AM to 10:28 AM. The background of the phone screen shows a wallpaper with cartoonish characters and a green theme.

TLP:CLEAR

Images Descriptions

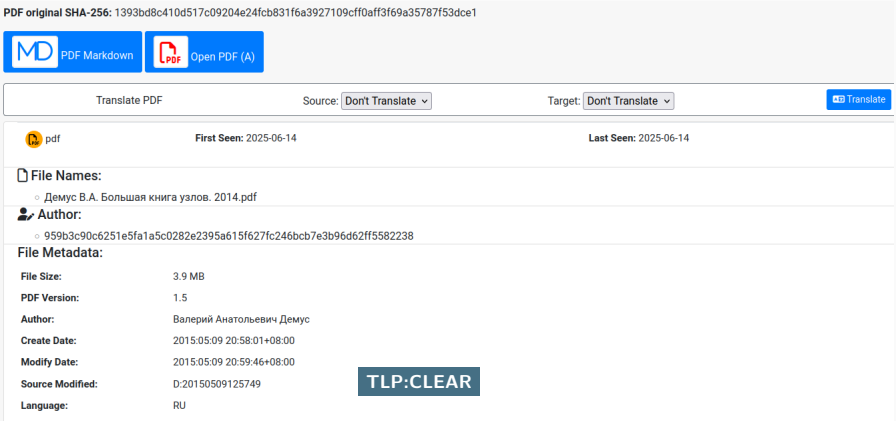
- Helps users quickly understand image content without viewing the image directly.
- Generated descriptions provide an extra layer of insight during analysis.



The image shows a black tactical scope mount. This type of mount is typically used for attaching a scope to a firearm, such as a rifle. The mount has a cylindrical opening in the center, which is designed to accept a scope. The mount also has several mounting points and adjustment screws, allowing for precise alignment and adjustment of the scope. The mount appears to be made of metal and is designed for durability and stability. The background shows a wooden surface with some packaging materials, indicating that the mount is new and possibly still in its original packaging.

PDFs

- Extracts PDF metadata and uses it as new pivotable data (author name).
- Converts PDFs to PDF/A and strips embedded metadata before ingestion to mitigate malicious payloads.
- Extracts content and processes it as Markdown.



TLP:CLEAR

PDF Translation

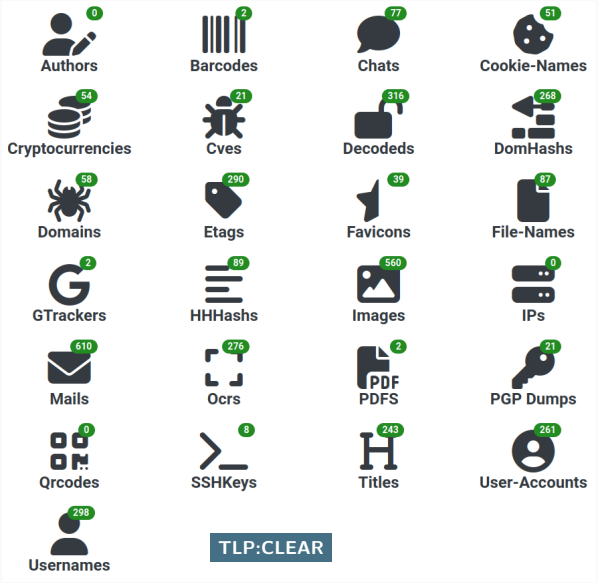
- Uses local translation (no external services required).
- Adds a translated text layer to the PDF while preserving the original content.



Tags Classification

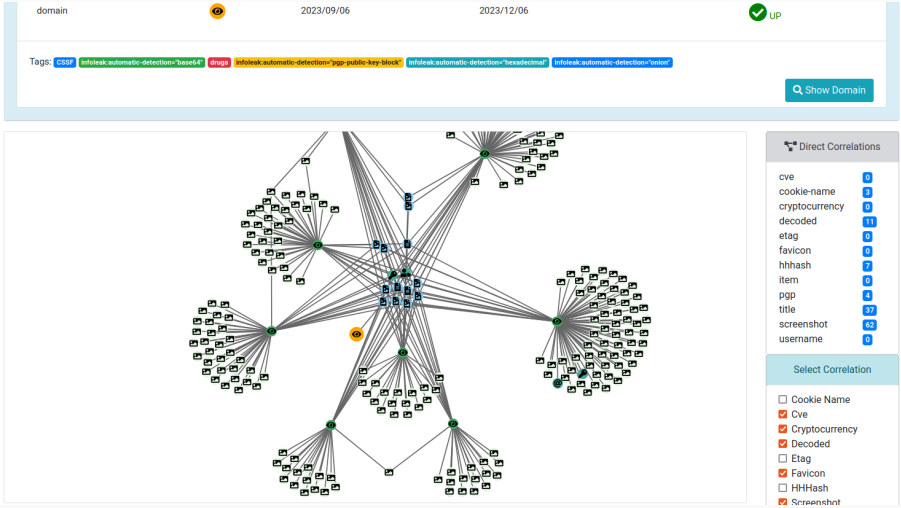
- Analysis modules automatically classify processed content
- Based on shared vocabularies (MISP taxonomies and galaxies)
- Automatic tags describe:
  - type of detected information (credential, credit-card, phone numbers, API key, ...)
  - origin of detection (automatic detection or analyst finding/validation)
  - source context (forum, website, repository, crawler)
- Enables searching and filtering by tags

Extracted and Correlated Objects

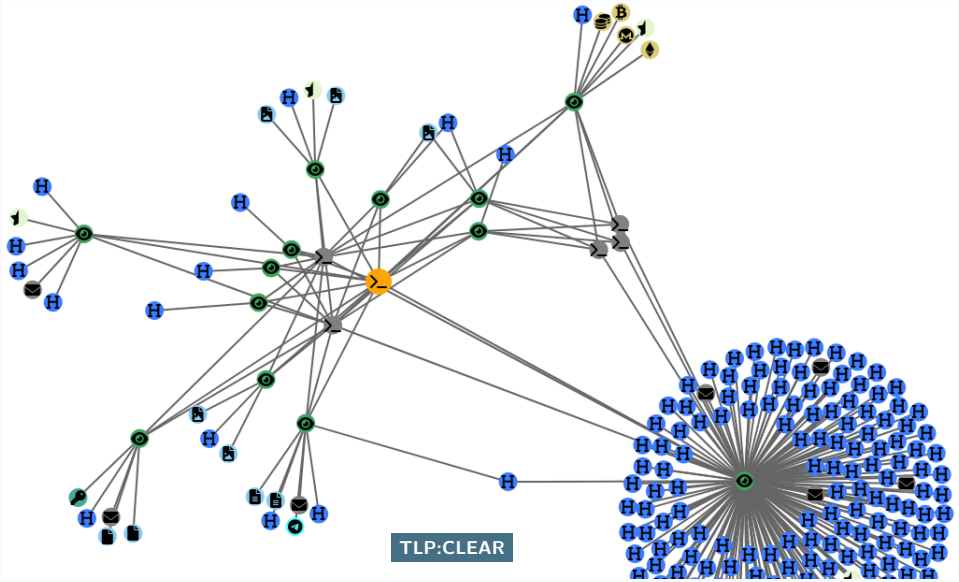


Correlation

Correlations - Favicon



Correlations - SSH Keys



TLP:CLEAR

Detection

Live Trackers

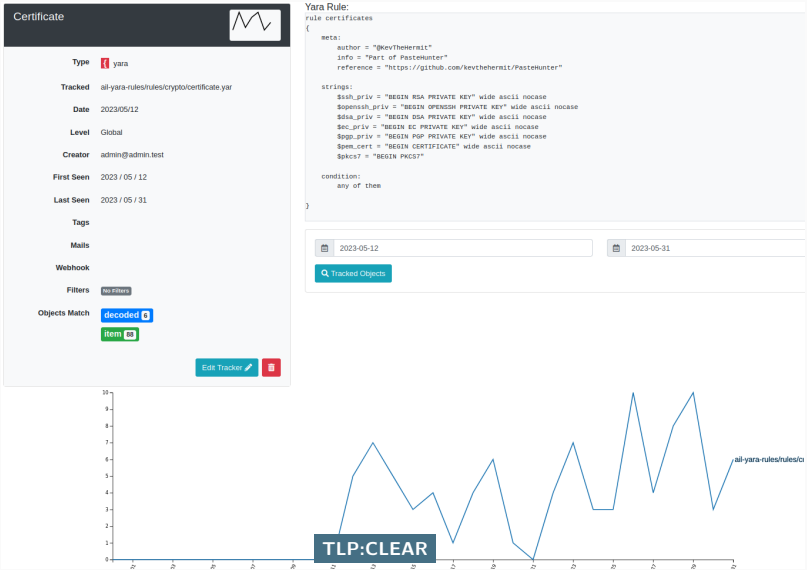
User-defined detection rules

- Live detection of collected data
- Real-time tagging and classification
- Email / Webhook notifications
- Matching objects are stored in trackers

Supported Detection Methods

- Track Word (exact keyword)
- Track Set (threshold-based keyword group)
- Regular expressions (Regex pattern matching)
- YARA rules (advanced structured detection)
- Typo-squatting domain generation

YARA Tracker



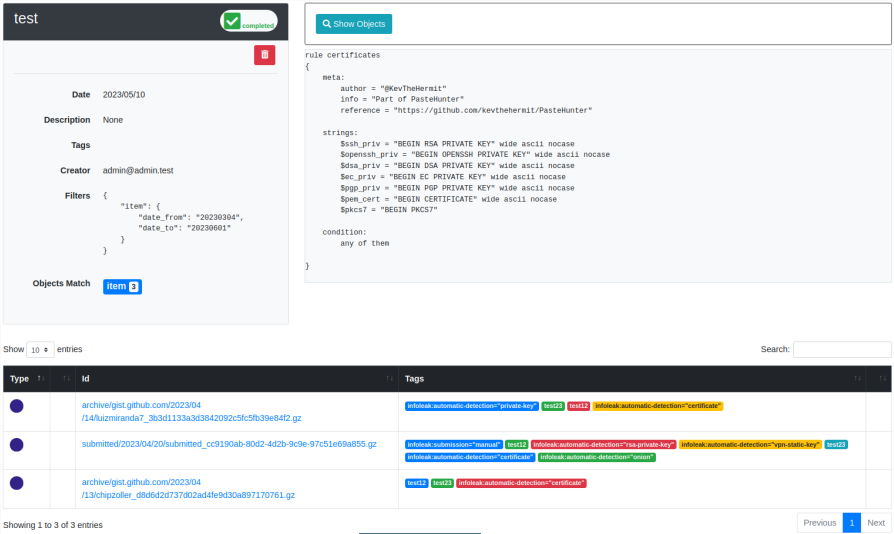
TLP:CLEAR

Retro Hunt

Applying detection rules to **Historical Data**

- Data is stored continuously by AIL
- A new **YARA detection rule** is created
- The YARA rule is executed against historical datasets
- Previously unnoticed content are identified

Retro Hunt



TLP:CLEAR

Creating a YARA Rule

What is YARA?

- An improved `grep` for pattern matching
- Searches for strings, binary patterns, and regular expressions
- Rules can include metadata and tags for context
- Easier and more structured conditions than complex regex

Basic Structure of a YARA Rule

- **rule** my_new_rule_name
- **meta:** descriptive information
- **strings:** patterns to search for
- **condition:** logical expression defining when it matches

YARA Rule Examples - Tracking a specific mail domain

```
rule mail_proton
{
    meta:
        description = "Search for a specific mail domain name"

    strings:
        $a = "@protonmail.com" nocase

    condition:
        $a
}
```

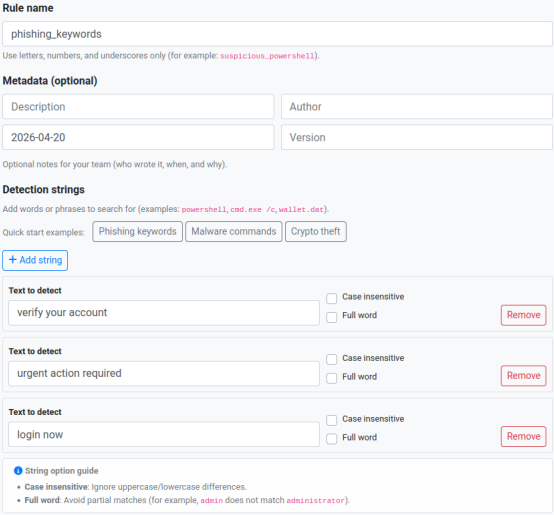
YARA Rule Examples - Detecting Disney+ credential leaks

```
rule disney_plus
{
    meta:
        description = "Finding list of credentials for Disney Plus"

    strings:
        $a = "gmail.com:" nocase
        $b = "DISNEY_PLUS"
        $c = "Disney Plus"

    condition:
        $a and ($b or $c)
}
```

YARA Rule Builder

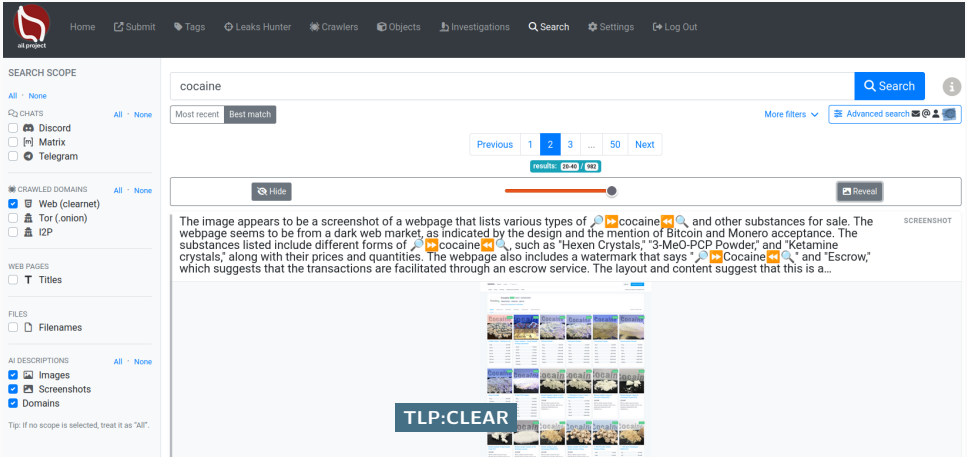


TLP:CLEAR

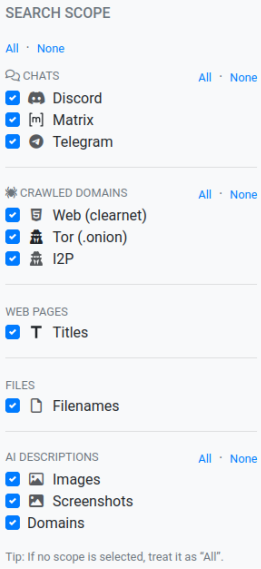
Search

Query Operators

- Use double quotes `"word"` for **exact matches** (disables fuzzy/typo matching)
- Use `-word` to **exclude terms** from results
- Use dot `.` as a **single-character wildcard**



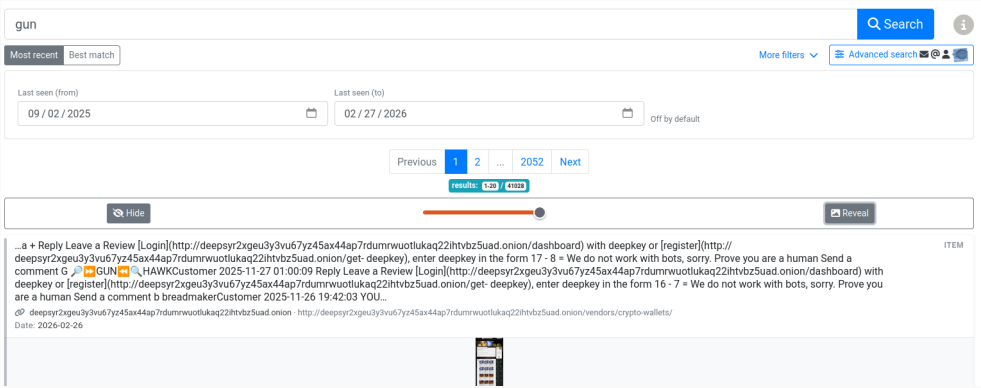
Search Scope



- Use the **left sidebar** to select what the query should search.
- Default: **All** sources (best for general discovery).
- Narrow scope to reduce noise and improve relevance:
  - **Chats**: Discord, Matrix, Telegram
  - **Crawled domains**: Web, Tor (.onion), I2P
  - **HTML titles**
  - **Filenames**
  - **AI descriptions**: Images, Screenshots, Domains

Result Ordering and Time Filters

- **Most Recent (default)**: Orders results by most recently seen objects first.
- **Recent**: Prioritizes results with stronger matches and higher occurrence counts.
- **Last Seen** Limits results to objects **last observed** within a specified date range.

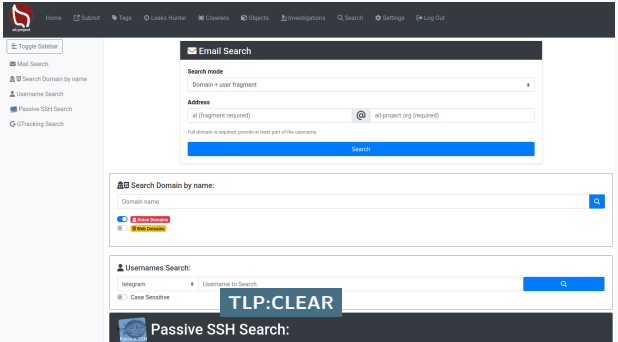


TLP:CLEAR

Advanced Search

Some searches require **structured input**:

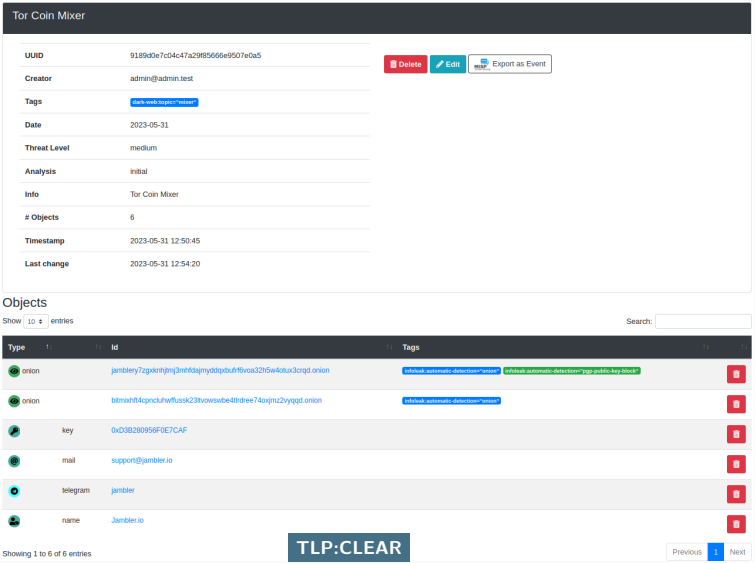
- **Email addresses**: exact match, user fragment, domain fragment
- **SSH keys**: fingerprints, key types, formatting
- **Usernames**:
  - If a username is used by an *user-account/chat*, it is already covered by the **Global Search**.
  - Usernames found in other sources (domains, QR codes, . . .) are **not currently indexed**.



TLP:CLEAR

Investigation

Investigations



TLP:CLEAR

Export





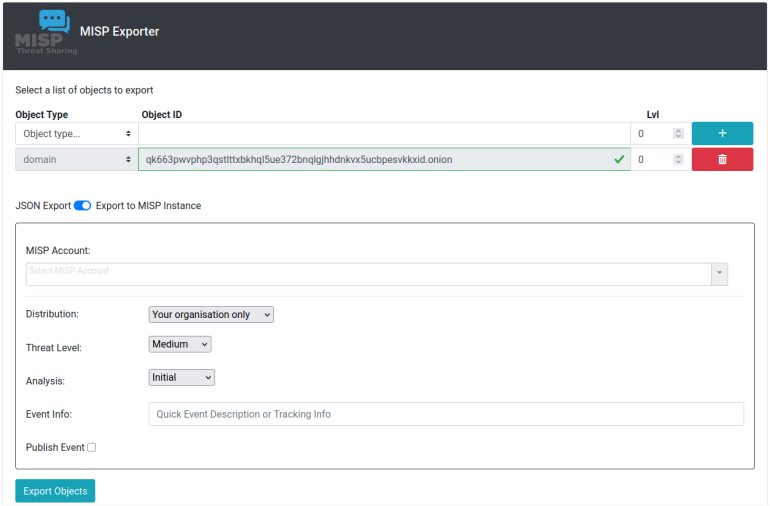


Detection modules are using the infoleak taxonomy[4]

1. Export AIL objects and investigations to MISP core format
2. Download them or Create a MISP Event[5]

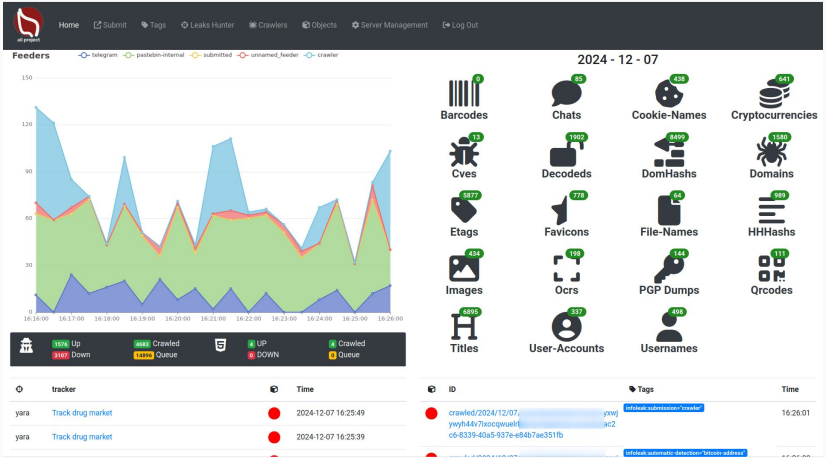
[4] https://www.misp-project.org/taxonomies.html
[5] https://www.misp-standard.org/rfc/misp-standard-core.txt

Objects Export



Live Demo

Dashboard



TLP:CLEAR

AIL Framework: Extensible Capabilities

- Extending AIL to add a new **analysis module** can be done in 50 lines of Python.
- The framework **supports multi-processors/cores by default**. Any analysis module can be started multiple times to support faster processing during peak times or bulk import.
- **Multiple** concurrent **data inputs**.
- Tor Crawler (handles cookies and authentication).
- Feeders: Discord, Telegram, ...

Ongoing Developments

- **Lacus crawler improvements:** remote headless features
- **Text geolocation**
  - **Improved pivoting:** search, cluster, and correlate data by place
  - **Geolocation tracker:** monitor specific geographic areas of interest
  - **visualization:** maps for objects, trackers and investigations
- **Advanced audio and video processing and extraction**

Links

- AIL project website `https://ail-project.org`
- AIL project open source framework `https://github.com/ail-project`
- Training materials `https://github.com/ail-project/ail-training`
- Online chat `https://gitter.im/ail-project/community`



MISP - LEA

- Law Enforcement Agency Information Sharing Community
- AIL-LEA instance hosted by CIRCL
- Request access by sending an email to **info@misp-lea.org** or **info@circl.lu**
- Request free training at **info@misp-lea.org** or **info@circl.lu**



Thank you for your attention

- AIL project[6] : `https://github.com/ail-project/ail-framework`
- For questions, contact: info@circl.lu

[6] All techniques and indicators mentioned in these slides are implemented in the AIL project, using an instance backed by a three-year dataset collected from Tor hidden services and various social networks.